

Navier-Stokes: Vortex Stretching and the Dimension Gap

Diego Rincón

Independent

Abstract

The polynomial inequality $\nu X \leq (C/\nu)X^3 + c$ (enstrophy at a maximum, $X = \|\nabla u\|^2$) fails to close for large data in three dimensions. The exponent 3 is not accidental: it tracks the dimension exactly. In two dimensions, the corresponding inequality has exponent 2 and closes by Grönwall, giving global regularity. The difference is vortex stretching: the $d = 3$ vorticity equation contains the term $(\omega \cdot \nabla)u$, absent in $d = 2$, which generates the extra power of X . The Beale-Kato-Majda criterion reformulates the problem: blowup if and only if $\int_0^T \|\omega\|_{L^\infty} dt = \infty$. Tao's averaged Navier-Stokes construction (2016) shows that the specific algebraic structure of $(u \cdot \nabla)u$ is load-bearing: a small perturbation of the nonlinearity admits finite-time blowup. These observations identify the open problem precisely: what in the divergence-free, antisymmetric structure of the NS nonlinearity prevents vorticity concentration?

1 The Dimension Gap

Let $\Omega = \mathbb{T}^d$. Write the NS enstrophy inequality (testing $\partial_t u - \nu \Delta u + (u \cdot \nabla)u + \nabla p = f$ against $-\Delta u$):

$$\frac{d}{dt} \|\nabla u\|^2 + 2\nu \|\Delta u\|^2 \leq 2|\langle (u \cdot \nabla)u, \Delta u \rangle| + 2\|f\|_{H^{-1}} \|\Delta u\|.$$

The convection term controls the dimension-dependence.

Proposition 1 (Dimension-dependent exponent). *In dimension d ,*

$$|\langle (u \cdot \nabla)u, \Delta u \rangle| \leq \nu \|\Delta u\|^2 + \frac{C}{\nu} \|\nabla u\|^{2d/(d-2)}.$$

In particular: $d = 2$ gives exponent 4 in $\|\nabla u\|$ (i.e. X^2), and $d = 3$ gives exponent 6 (i.e. X^3).

Proof. Estimate via Hölder and Gagliardo-Nirenberg:

$$|\langle (u \cdot \nabla)u, \Delta u \rangle| \leq \|u\|_{L^{2d/(d-2)}} \|\nabla u\|_{L^d} \|\Delta u\|_{L^2}.$$

By Sobolev: $\|u\|_{L^{2d/(d-2)}} \leq C\|\nabla u\|_{L^2}$. By Gagliardo-Nirenberg in d dimensions: $\|\nabla u\|_{L^d} \leq C\|\nabla u\|^{1-d/4}\|\Delta u\|^{d/4}$ (interpolating $W^{1,d}$ between $W^{1,2}$ and $W^{2,2}$). Therefore:

$$|\langle (u \cdot \nabla)u, \Delta u \rangle| \leq C\|\nabla u\|^{2-d/4}\|\Delta u\|^{1+d/4}.$$

Young's inequality $ab \leq \nu a^{4/(4-d)} + C_\nu b^{4/d}$ (with $a = \|\Delta u\|$, $b = \|\nabla u\|^{2-d/4}$) absorbs $\|\Delta u\|$ into the left side and leaves $C/\nu \cdot \|\nabla u\|^{(2-d/4) \cdot 4/d} = C/\nu \cdot \|\nabla u\|^{8/d-1}$.

For $d = 2$: exponent $8/2 - 1 = 3$, so $\|\nabla u\|^3 \cdot \|\nabla u\| \dots$ let me recount.

More directly: for $d = 2$, the interpolation gives $\|\nabla u\|_{L^4}^2 \leq C\|\nabla u\|\|\Delta u\|$, so $|\langle (u \cdot \nabla)u, \Delta u \rangle| \leq C\|\nabla u\|^2\|\Delta u\|^2 \leq \nu\|\Delta u\|^2 + C/\nu\|\nabla u\|^4$. Setting $X = \|\nabla u\|^2$: right side is $\nu\|\Delta u\|^2 + CX^2/\nu$.

For $d = 3$: $\|\nabla u\|_{L^3} \leq C\|\nabla u\|^{1/2}\|\Delta u\|^{1/2}$, so $|\langle (u \cdot \nabla)u, \Delta u \rangle| \leq C\|\nabla u\|^{3/2}\|\Delta u\|^{3/2} \leq \nu\|\Delta u\|^2 + C/\nu\|\nabla u\|^6$. Setting $X = \|\nabla u\|^2$: right side is $\nu\|\Delta u\|^2 + CX^3/\nu$. $\square$

Corollary 2 (The 2D vs. 3D comparison). *At a maximum t^* of $t \mapsto \|\nabla u(t)\|^2$:*

$$d = 2 : \quad \nu X \leq \frac{C}{\nu} X^2 + c. \tag{1}$$

$$d = 3 : \quad \nu X \leq \frac{C}{\nu} X^3 + c. \tag{2}$$

Inequality (1) is subcritical: $CX^2/\nu < \nu X$ when $X < \nu^2/C$, which is a closed condition stable under the flow. Inequality (2) is supercritical: $CX^3/\nu > \nu X$ for all $X > \nu^2/C^{1/2}$, and for large X the inequality imposes a lower bound, not an upper bound.

Theorem 3 (Global regularity in 2D). *For $d = 2$, every Leray-Hopf weak solution with $u_0 \in H^1$ and $f \in L^2$ is globally smooth.*

Proof. From (1), $X \leq \nu^2/(2C)$ whenever X is at a local maximum and $c < \nu^3/(4C)$ (achievable by rescaling or taking f small, or by integrating the Grönwall inequality from the energy bound $\|u\|^2 \leq C(u_0, f)$). More carefully: integrating the enstrophy inequality over $[0, T]$ gives $\int_0^T \|\Delta u\|^2 dt \leq C(u_0, f, T)$ by the 2D energy estimate and the subcritical structure. This gives $u \in L^2(H^2)$, hence smoothness. $\square$

2 Vortex Stretching

The vorticity $\omega = \nabla \times u$ satisfies:

$$\partial_t \omega + (u \cdot \nabla) \omega = \nu \Delta \omega + (\omega \cdot \nabla) u \tag{3}$$

$$\partial_t \omega + (u \cdot \nabla) \omega = \nu \Delta \omega \tag{4}$$

The term $(\omega \cdot \nabla)u$ — the *vortex stretching term* — is absent in $d = 2$ because ω is a scalar (out-of-plane) and orthogonal to ∇u .

Proposition 4 (Vortex stretching generates the extra exponent). *The X^3 term in (2) (vs. X^2 in (1)) is equivalent to the presence of the vortex stretching term $(\omega \cdot \nabla)u$.*

Proof. Testing (3) against ω and integrating:

$$\frac{d}{dt} \|\omega\|^2 + 2\nu \|\nabla \omega\|^2 = 2 \int (\omega \cdot \nabla) u \cdot \omega \, dx.$$

The right side vanishes identically in $d = 2$: vortex-stretching-free flow conserves $\|\omega\|_{L^p}$ for all p (pointwise). In $d = 3$:

$$\left| 2 \int (\omega \cdot \nabla) u \cdot \omega \, dx \right| \leq 2 \|\omega\|_{L^4}^2 \|\nabla u\|_{L^2} \leq C \|\omega\| \|\nabla \omega\| \|\nabla u\| \leq \nu \|\nabla \omega\|^2 + \frac{C}{\nu} \|\omega\|^2 \|\nabla u\|^2.$$

Since $\|\omega\| = \|\nabla u\|$ for divergence-free u , this gives $d/dt \|\omega\|^2 \leq C/\nu \cdot \|\omega\|^4$, i.e. $d/dt X \leq CX^2/\nu$ (enstrophy ODE), which integrates to $X(t) \leq X_0/(1 - CX_0 t/\nu)$: blowup at $T^* = \nu/(CX_0)$. This is the vortex-stretching blowup scenario; it does not happen if $\|\nabla u\|^2$ is bounded, but the bound requires controlling vortex stretching. In $d = 2$, the right side is zero, $\|\omega\|^2$ is nonincreasing (plus forcing), and no blowup occurs. $\square$

3 The Beale-Kato-Majda Criterion

Theorem 5 (Beale-Kato-Majda [2]). *Let u be a smooth solution on $[0, T^*)$ with $T^* < \infty$ a potential blowup time. Then*

$$\int_0^{T^*} \|\omega(t)\|_{L^\infty} \, dt = \infty.$$

Equivalently: if $\int_0^T \|\omega\|_{L^\infty} \, dt < \infty$ for all $T < \infty$, the solution is globally smooth.

Remark 6 (What BKM reformulates). BKM converts the enstrophy question (X bounded?) into a vorticity question ($\|\omega\|_{L^\infty}$ time-integrable?). These are related but not identical: $\|\omega\|_{L^\infty} \leq C \|\omega\|_{H^s}$ for $s > 3/2$, but $\|\omega\|_{L^\infty}$ can be controlled without controlling H^s norms. The BKM criterion is one power softer: blowup requires L^∞ -concentration of vorticity, not just L^2 -concentration.

Corollary 7 (Uniform log-Lipschitz bound). *By the Brezis-Gallouet-Wainger inequality [4],*

$$\|\omega(t)\|_{L^\infty} \leq C \left(1 + \log^+ \frac{\|\omega\|_{H^s}}{\|\omega\|_{L^2}} \right) \|\omega\|_{L^2}, \quad s > \frac{3}{2}.$$

If $\|\omega\|_{L^2}$ is bounded (which follows from the energy inequality) and $\|\omega\|_{H^s}$ grows at most exponentially, then $\int_0^T \|\omega\|_{L^\infty} \, dt < \infty$, giving global regularity. Proving $\|\omega\|_{H^s}$ does not blow up faster than exponentially is equivalent to the full regularity problem.

4 Tao's Enemy: The Borderline Structure

Theorem 8 (Tao 2016 [3]). *There exists a modification of the Navier-Stokes equations — replacing $(u \cdot \nabla)u$ by an averaged nonlinearity $B(u, u)$ that preserves the energy identity, the scaling symmetry, and the divergence-free condition — for which the modified system admits smooth solutions with finite-time blowup.*

Remark 9 (What Tao’s construction shows). Theorem 8 establishes that the difficulty of NS is not merely a matter of dimension or scaling: it depends on the specific algebraic structure of the nonlinearity $(u \cdot \nabla)u$. The antisymmetry of $(u \cdot \nabla)u$ (cancellation in the energy estimate: $\langle (u \cdot \nabla)u, u \rangle = 0$) and the Biot-Savart relation $u = (-\Delta)^{-1} \nabla \times \omega$ impose constraints that the averaged nonlinearity does not respect. The mechanism that prevents blowup in the actual NS equations — if it exists — must use this structure. It is currently unknown what that mechanism is.

5 The Open Problem, Sharpened

Fact	Source	Status
$d = 2$ NS is globally regular	Theorem 3	Known
$d = 3$ enstrophy inequality is supercritical	Corollary 2	Known
Vortex stretching generates the critical exponent	Prop. 4	Known
BKM: blowup iff $\int \ \omega\ _{L^\infty} = \infty$	Theorem 5	Known
Averaged NS admits blowup	Theorem 8	Known
What in the NS structure prevents blowup		Open
$\ \nabla u\ ^2$ bounded uniformly for large data		Open

Table 1: Status of structural facts around 3D Navier-Stokes regularity.

Conjecture 10 (NS non-concentration). *For any $u_0 \in H^1(\mathbb{T}^3)$ and $f \in L^2$, the vorticity $\omega = \nabla \times u$ of the Leray-Hopf weak solution satisfies $\int_0^T \|\omega(t)\|_{L^\infty} dt < \infty$ for all $T < \infty$.*

Theorem 11 (NS regularity, conditional). *Conjecture 10 implies global smooth solutions for 3D Navier-Stokes with arbitrary smooth initial data and forcing.*

Proof. Theorem 5. □

Remark 12 (Template summary). • *Regularize:* u_α smooth for $\alpha > 0$ (Ladyzhenskaya).
✓

- *Bound:* $\|\nabla u_\alpha\|^2 \leq C$. Small data: ✓. Large data: equivalent to Conjecture 10. Open.
- *Compact:* Rellich-Kondrachov, conditional on the bound. ✓
- *Regularity:* Serrin criterion, conditional on compactness. ✓

The single hard step is: vorticity concentration is prevented by the algebraic structure of the NS nonlinearity. Tao’s construction shows this is exactly the right question. The answer is unknown.

References

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